

curve $(\rho_t)_{t \in [0,1]}$ is a geodesic

Displacement Interpolation

- If an OT map T exists between α to β , the optimal path is given by

$$\rho_t = \left[(1 - t)\text{Id} + tT \right]_{\#} \alpha$$

displacement /
McCann interpolation

- The corresponding vector field is $v_t(x) = T(x) - x$
- Each datapoint/particle follows straight line: $x_t = (1 - t)x + tT(x)$
- The **curve** $(\rho_t)_{t \in [0,1]}$ **is a geodesic** in Wasserstein space
- If no OT map exists, we can still define displacement with coupling π :

$$\rho_t = \left[(1 - t)x + ty \right]_{\#} \pi$$

Displacement vs Mixture Interpolation

The simplest alternative to McCann is the ‘naive’ **mixture interpolation**

Displacement Interpolation:

$$\rho_t^{\text{disp}} = \left[(1 - t)\text{Id} + tT \right]_{\#} \alpha$$